

Sun Yat-sen University

undergraduatecoursesyllabus

College (Department)School of Mathematics

Course Name: Linear Algebra

2025

coursesyllabus

(Date of Writing: September 2025)

1. Basic information of the course

Linear Algebra Linear Algebra					
coursecategory	Special	Course Code	MA179	Course Offering Unit	School of Mathematics
credit	3	Study Hours	54	Teaching grade	Freshman or Sophomore
For majors/categories	Economics, Management, Science, Engineering, Agriculture				
Course Director	Suning				
Prerequisite Courses	none				
Course Objectives	Linear algebra is essential mathematical knowledge for science and engineering students. This course requires students to gain a firm grasp of fundamentals such as solving systems of linear equations, calculating and applying determinants, matrix operations, elementary row transformations, rank, matrix eigenvalues, eigenvectors, and similarity diagonalization. Students should also develop a general understanding of block matrices, quadratic form simplification, positive definite matrices, and linear spaces. Students should also gain a general understanding of Jordan's standard form and linear transformations. This course lays the foundation for students to further study other subsequent mathematics courses and specialized programs. It also enhances students' logical thinking, analytical, and problem-solving abilities. In terms of ideological and political education, students are guided to understand that linear algebra stems from and serves practice, thereby fostering a dialectical materialist worldview and cultivating good learning habits, mathematical literacy, and a truth-seeking attitude.				

2. Basic content of the course (requires a certain word count and cannot be too simple)

(1) Teaching schedule

Chapter order and name (detailed to section, required)	Main teaching content (required)	Required study hours (required)	Educational elements	Key points and difficulties (optional)	Week (optional)	Remarks (optional)
Chapter 1 Linear Equations 1. Gauss-Jordan algorithm	Definition of linear equations; The definition of matrix and how to use matrix to represent linear equations (coefficient matrix, augmented matrix); Elementary row transformations of matrices; Definition of echelon matrix; Gauss-Jordan algorithm for solving linear equations	2	This section introduces the ancient Chinese mathematical treatise "Nine Chapters on the Mathematical Art." The book uses the method of separating coefficients to represent linear equations, equivalent to the modern coefficient matrix. Its method for solving linear equations is essentially the same as the Gauss-Jordan algorithm, a similar approach not systematically proposed in the West until the 17th century.	Key Points: Gauss-Jordan Algorithm		
Chapter 1 Linear Equations 2. Solutions of linear equations and their judgment criteria	Definition of homogeneous and nonhomogeneous linear equations; Determine the existence and uniqueness of solutions to linear systems; in particular, determine whether homogeneous linear systems have nonzero solutions; Briefly introduce the definition of number field (not an assessment requirement).	2	This course introduces the origin of the Chinese word "algebra," which was translated by Qing Dynasty mathematician Li Shanlan. By introducing Li Shanlan's contributions, students can appreciate the exploration process of modern Chinese mathematicians.	Key Points: Determination of the Existence and Uniqueness of Solutions to Linear Equations		The number field in this course is assumed to be the real number field (unless otherwise specified).
Chapter 2 Determinants 1. Definition of n-ary permutations and determinants	Introduce the definition of inverse numbers; Introducing the complete expansion of n-order determinants; Methods for finding the determinants of upper and lower triangular matrices.	2	none	Difficulty: Complete expansion of the determinant		
Chapter 2 Determinants 2. Properties of determinants and expansion of determinants into rows (columns)	The impact of elementary row and column transformations on determinants, and how to use row (column) transformations to calculate determinants; Transposing does not change the determinant; Cofactor, definition of algebraic cofactor; The rule for expanding	2	The calculation of determinants embodies the idea of "analyzing specific issues in specific ways"	Difficulty: Calculation of determinants		The proof of the properties of the determinant can be skipped

	determinants by rows (columns) and how to use this rule to calculate determinants.					
Chapter 2 Determinants 3. Cramer's Rule	A system of linear equations with a square coefficient matrix has a unique solution if and only if the determinant of the coefficient matrix is not equal to zero; Cramer's rule; The determinant is expanded into k rows (columns) (not a requirement for the assessment).	2	The use of determinants to determine whether a linear equation has a unique solution was first proposed in the "Nine Chapters on the Mathematical Art", which is the contribution of ancient Chinese mathematics to algebra.	Key Point: The relationship between the unique solution of a linear equation and the determinant		
Chapter 3 n-Dimensional Vector Space 1. n-dimensional vector space and its subspaces	Definition of n-dimensional vector space; Row vectors and column vectors; Linear combination of vector groups, linear representation; The relationship between the linear representation of a vector group and a linear equation system; Definition of subspace; The space generated by the group of vectors.	2	(If you have time) You can introduce the definition of a general vector space. This embodies the idea of "axiomatization" in modern mathematics.	Key Points: Definition of vector space and subspace; The relationship between the linear representation of a vector group and a linear equation group		
Chapter 3 n-Dimensional Vector Space 2. Linearly dependent and linearly independent vector groups	Definition of linear dependence and linear independence; The relationship between linear dependence (independence) and linear combination; The relationship between linear dependence (independence) and homogeneous linear equations; How to determine whether a vector group is linearly dependent (independent)? Extension group and shortening group.	2	none	Key Points: The relationship between linear dependence (independence) and linear combination; The relationship between linear dependence (independence) and homogeneous linear equations		
Chapter 3 n-Dimensional Vector Space 3. Maximal linearly independent groups and the rank of vector groups, basis and dimension of	Definition of maximal linearly independent groups; Equivalence of vector groups; Define the rank of a vector group and prove that equivalent vector groups have the same rank; Definition of basis and dimension of vector	2	From a basis of a vector space, we can uniquely represent any vector in that space. This demonstrates the dialectical relationship between "finite" and "infinite." It also teaches us that when solving problems,	Key point: Any two bases of a vector space contain the same number of vectors		

vector spaces	space; Definition of coordinates; The column space and row space of a matrix.		grasping the key points can simplify complex problems.			
Chapter 3 n-Dimensional Vector Space 4. Matrix rank, necessary and sufficient conditions for linear equations to have solutions	Definition of row rank and column rank of a matrix, and proof that the row rank of a matrix is equal to the column rank; How to find the maximal linearly independent group through row transformation; Transposing does not change the rank of the matrix; The relationship between the rank of a matrix and its minors; Obtaining the necessary and sufficient conditions for the solution of linear equations from the perspective of rank	2	As long as the number of vectors in a linearly independent group reaches the dimension of the vector space, the linearly independent group becomes a basis equal to its column rank, which embodies the idea of "quantitative change leads to qualitative change".	Key Point: The row rank of a matrix is equal to its column rank		
Chapter 3 n-Dimensional Vector Space 5. The structure of solutions to linear equations (both homogeneous and inhomogeneous)	The solution set of a homogeneous linear system of equations is a vector space; Basic solution system of homogeneous linear equations; The relationship between the dimension of the solution space and the rank of the coefficient matrix; The relationship between the solution set of a non-homogeneous linear system and the solution set of the corresponding homogeneous linear system;	2	none	Key Points: The relationship between the solution set of a non-homogeneous linear system and the solution set of the corresponding homogeneous linear system; Basic properties of matrix operations (especially the associative and distributive properties of multiplication); How to use elementary matrix multiplication to achieve elementary transformations	Antisymmetric matrices only need to be understood and are not included in the assessment scope.	
Chapter 4 Matrix Operations 1. Definition of matrix operations and special matrices	Definition and basic properties of matrix addition, scalar product, multiplication, and transposition; Represent linear equations using matrix multiplication; Definitions of diagonal matrices, upper (lower) triangular matrices, elementary matrices, symmetric matrices, and antisymmetric matrices; The relationship between elementary matrices	2	None			

	elementary row (column) transformations					
Chapter 4 Matrix Operations 2. Rank and determinant of matrix products, reversible matrices, and partitioning	The relationship between $\text{rank}(AB)$ and $\text{rank}(A)$ and $\text{rank}(B)$; The determinant of AB is equal to the determinant of A times the determinant of B; Binet-Cauchy formula (optional, not required for assessment); The definition of a reversible matrix, and some necessary and sufficient conditions for a matrix to be reversible; The definition of adjoint matrix and how to use it to calculate reversible matrices, the inverse matrix; Elementary transformation methods for finding the inverse matrix; Solve matrix equations using inverse matrices; Block diagonal matrices and block upper (lower) triangular matrices; Elementary row transformations of block matrices (optional, not required for assessment); Block matrix multiplication	4	(If you have time) You can introduce the Leontiev input-output model. This model uses matrix operations to study the relationship between input and output in economics, and reflects the important role of linear algebra in macroeconomics.	Key Points: The relationship between $\text{rank}(AB)$ and $\text{rank}(A)$ and $\text{rank}(B)$; Various necessary and sufficient conditions for reversible matrices		
Chapter 4 Matrix Operations 3. Orthogonal matrices, Euclidean space, linear mapping from K^n to K^s	The definition of orthogonal matrices and their basic properties; columns of an orthogonal matrix; Standard inner product on R^n; Definitions of orthogonal vector groups, orthogonal bases, and orthonormal bases; Schmidt orthogonalization process; Definition of linear mapping; The kernel and image of a linear map, as well as the dimensional formula of a linear map (not required for assessment)	4	Through the study of orthogonal relations and orthogonal matrices, students can understand the idea of "combining numbers and shapes".	Key Points: Basic properties of orthogonal matrices; Definition of inner product, Schmidt orthogonalization process;		If you have time, you can further explore the relationship between linear mapping and matrix, and prove that every linear mapping can be "represented" by a matrix.
	Introduce the definition of matrix cancellation and prove					

Chapter 5: Comparison and Similarity of Matrices 1. Definition and basic properties of matrix offset and similarity, and the definition of eigenvalues and eigenvectors	that two matrices cancel if and only if their ranks are equal (this property is not a requirement for the exam); Introduce the definition of generalized inverse and its applications (generalized inverse is not a requirement for the exam); Similar definition of matrix; Prove that similar matrices have the same determinant, trace, and rank; Definition of diagonalizable matrix; Definition of eigenvalues and eigenvectors	4	The offset/similarity of matrices embodies the idea of "seeing the essence through the phenomenon"	Key Points: The definition of matrix similarity and the properties that two similar matrices have in common; Definition of eigenvalues and eigenvectors		
Chapter 5: Comparison and Similarity of Matrices 2. How to find the eigenvalues and eigenvectors of a matrix, and the necessary and sufficient conditions for diagonalization of the matrix	characteristic polynomial; How to find the eigenvalues of a matrix; The definition of characteristic subspace and how to find a set of bases for characteristic subspace; The coefficients in the characteristic polynomial (the proof process can be omitted); Necessary and sufficient conditions for matrix diagonalization; The eigenvectors corresponding to different eigenvalues are linearly independent;	4	We can also introduce Arthur Cayley, the mathematician and one of the founders of matrix theory. Cayley not only made significant contributions to the development of algebra, but also actively promoted educational equality for both sexes.	Key Points: How to find the eigenvalues and eigenvectors of a matrix; Necessary and sufficient conditions for matrix diagonalization		
Chapter 5: Matrix Compatibility and Similarity 3. Diagonalization of Real Symmetric Matrices	Definition of orthogonal similarity; The eigenvectors of a real symmetric matrix corresponding to different eigenvalues are orthogonal; Orthogonal diagonalization of real symmetric matrices;	2	none	Key Points: Orthogonal diagonalization of real symmetric matrices;		
Chapter 6 Quadratic Forms, Matrix Contracts 1. Quadratic form and its standard form	Definition of quadratic form; The relationship between quadratic forms and symmetric matrices; Definition of the standard form of a	2	none	Key Points: The relationship between quadratic forms and symmetric matrices		

	quadratic form					
Chapter 6 Quadratic Forms, Matrix Contracts	Nondegenerate linear substitutions and orthogonal substitutions in quadratic forms; Equivalence of quadratic forms and contraction of matrices; The standard form of the quadratic form; Methods for matching standard forms (optional, not required for assessment); Normal form of real symmetric matrices, and positive (negative) inertia indices; Inertia Theorem (proof not required)	4	Through the discussion of the relationship between the standard form of quadratic form and the classification of quadratic surfaces, students can further deepen their understanding of "combining numbers and shapes"	Key Points: Convert the quadratic form into standard form through orthogonal substitution; Uniqueness of canonical form		
Chapter 6 Quadratic Forms, Matrix Contracts	The definition of positive definite quadratic forms, and the relationship between positive definite quadratic forms and positive inertia indices; The definition of a positive definite matrix, and the relationship between a positive definite matrix and its eigenvalues; Use the sequential principal minor to determine the positive definiteness of the matrix; Introducing other types of quadratic forms	4	Through the application of positive (negative) definite matrices in extreme value problems, we can understand the connection between multivariate calculus and linear algebra.	Key Points: Several necessary and sufficient conditions for determining a positive definite matrix (positive definite form) (positive inertia index, eigenvalue, and sequential principal minor)		
Check for deficiencies and fill in the gaps Course Review	Check and fill in the gaps in the previous content; Help students sort out the overall context of the course; Deepen students' understanding and grasp of the key and difficult points of the course through reviewing and explaining exercises	4	none			

Total: 54

(2) Teaching arrangements

(Necessary explanations should be given for the arrangement of various teaching links, such as experiments, internships, exercises, homework, etc., as well as the connection and division of labor between this course and other related courses. The arrangement of teaching links should reflect high-level, innovative and challenging characteristics.)

Homework: due once a week; Exercise class: exercises for each chapter will be explained at the end of each chapter, or arranged by the instructor as appropriate.

(3) Teaching methods

(This includes classroom lectures, discussion sessions, after-class exercises and Q&A sessions. It is important to increase team learning, group assignments, the integration of experimental and theoretical classes, the use of information technology, and the shift from a teacher-

centered and knowledge-centered approach to a student-centered and learning-centered approach.)

Classroom teaching: blackboard writing combined with PowerPoint. Question and answer: one class per week is used for question and answer and after-class discussion (to be arranged by the teacher).

Optional: Encourage students to use AI tools such as large language models to deepen their understanding of concepts and explore the application of linear algebra in their majors. At the same time, emphasize to students the core role of independent thinking in course learning.

(IV) Course Textbooks (The main teaching materials should be those from the "Marxism Project" and national planning textbooks as much as possible. Among similar textbooks, national planning textbooks are preferred. If there are key textbooks from the "Marxism Project" in the textbook selection range, key textbooks from the project must be used.)

"Advanced Algebra" (4th edition, Volume 1), Qiu Weisheng, Higher Education Press.

(V) Main reference books

(Recommend several reference books and indicate the title, author, publisher, version, publication date, etc., and specify a certain amount of clear reading materials for each chapter)

1. Linear Algebra, edited by the Department of Mathematics of Tongji University, published by Higher Education Press.

2. Linear Algebra, Qi Minyou, Higher Education Press

3. Advanced Algebra, edited by the Department of Mathematics, Peking University, Higher Education Press

(6) Method of Assessment

Course assessment should adhere to the combination of process-oriented and result-oriented assessment, adopt multiple forms and multiple stages of assessment methods, and regular scores generally account for about 40% to 60%.

Assessment method: Examination

Score entry method: Percentage

1. Composition of the total score: Final examination accounts for 60%, and regular examination accounts for 40%.

2. Final Examination Instructions:

The final exam is a closed-book written exam, scored on a 100-point scale. The score for this section is calculated by multiplying the score by 0.6. The exam covers all chapters (unless otherwise noted), with scores for each chapter being roughly evenly distributed. Question types include true/false questions, fill-in-the-blank questions, calculation questions, and proof questions.

III. Description of routine assessment:

1. Class participation and attendance (accounting for 10% of the total score). Leave must be requested in accordance with school regulations and must be requested in advance. Leave cannot be requested by others on behalf of the applicant. If leave is not requested in advance due to special circumstances, the applicant must complete the application within one week and attach an explanation of the circumstances.

Otherwise it will be treated as absence.

2. Homework (20% of the total grade). Homework is generally assigned once a week. The main part of the homework is the exercises in the textbook and will be reviewed by the teacher.

Assignments must be submitted upon return from leave, otherwise they will be treated as unsubmitted assignments.

3. In-class quizzes (10% of the total score). The timing and format of in-class quizzes are determined by the instructor.

Note: The syllabus should be on A4 paper, and the text should be in size 12-point Song font.

Sun Yat-sen University

undergraduatecoursesyllabus

College (Department)Lingnan College

Course Name Microeconomics

2025

coursesyllabus

(Date of Writing: July 2025)

1. Basic information of the course

Microeconomics Microeconomics					
coursecategory	Special	Course Code	LN133	Course Offering Unit	Lingnan College
credit	3	Study Hours	54	Teaching grade	1
For majors/categories	Economics, Economics Major, Economics, Economics, Management Major, Management Science Engineering, Tourism Management, Business Administration, Business Management, Management Science, Tourism Management, Tourism Management, Management Science, New Major in Business Administration				
Course Director	Cai Guowei				
Prerequisite Courses	none				
Course Objectives	Course Objectives 1. Provide students with basic knowledge on consumption, production and market structure; 2. Reveal and cultivate students' basic skills in applying data and charts in economic analysis; 3. Guide students to extract basic concepts from economic environment and business activities; 4. Develop students' creative thinking of optimization, balance and interaction; 5. Inspire students to apply basic concepts and methods of microeconomics to analyze China's current economic problems. Basic requirements: 1. Have a firm grasp of the basic theories of consumption, production, and market structure;				

	2. Be able to skillfully use charts and mathematical tools to depict economic phenomena; 3. Be able to flexibly apply basic concepts to analyze China's current economic problems and understand China's economic policies. Ideological and political education in courses: Integrate Chinese practices, Chinese cases and Chinese stories into classroom teaching content, inspire students to use the basic principles of Marxism to understand the applicability of modern microeconomics in the development of China and the world economy, and gradually introduce new developments in Chinese microeconomics into classroom teaching.
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2. Basic content of the course (requires a certain word count and cannot be too simple)

(1) Teaching schedule

Chapter order and name (detailed to section, required)	Main teaching content (required)	Required study hours (required)	Educational elements	Key points and difficulties (optional)	Week (optional)	Remarks (optional)
Chapter 1: Core Concepts in Economics	Core Concepts in Economics	2	Explain basic economic concepts in combination with the concept of socialist market economy with Chinese characteristics	Key points: What is economics and the importance of economics		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 2: Modern Mixed Economy	Modern mixed economy	2	Based on China's reality, the author explains the respective advantages and roles of the government and the market, and by comparing them with the situations in other countries, explains the characteristics of the modern mixed economy.	Focus: Introduce the characteristics of major economic systems		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 3: The Fundamentals of Supply and Demand	The basic principles of supply and demand	2	This paper discusses the characteristics of the supply side and the demand side in light of China's reality, and briefly introduces the situation and results of China's supply-side reform.	Key points: Basic concepts of supply, demand, and equilibrium; Difficulty: Distinguishing between supply and quantity, and demand and quantity		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 4: Supply and Demand: Elasticity and Applications	Supply and Demand: Elasticity and Applications	4	Combined with China's reality, for example, support policies for agriculture.	Key point: The concept of elasticity Difficulty: The economic meaning and mathematical expression of elasticity		Adjust the course content and progress due to the actual situation of military training and holidays
Exercises: First 4 Chapters	Explanation of exercises in the first 4 chapters	2	Practice questions are written using Chinese cases in light of China's reality.			Adjust the course content and progress due to the actual situation of military training and holidays

Chapter 5: Demand and Consumer Behavior	Demand and consumer behavior	4	Based on China's actual situation, analyze my country's consumer behavior decisions.	Key points: The concept of utility, derivation of demand curve Difficulty: Substitution effect and income effect		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 6: Production and Business Organization	Production and commercial organization	2	In light of China's reality, we will use large Chinese enterprises as examples.	Key points: production function, economies of scale, long-term and short-term Difficulty: Total Factor Productivity		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 7: Cost Analysis	Cost Analysis	4	Combined with China's reality, such as the development and current situation of China's steel enterprises.	Key points: The concepts of fixed cost, variable cost, and opportunity cost Difficulty: Economic substitution rate, technological substitution rate		Adjust the course content and progress due to the actual situation of military training and holidays
Exercises: Chapters 5-7	Explanation of Exercises 5-7	2	Practice questions are written using Chinese cases in light of China's reality.			Adjust the course content and progress due to the actual situation of military training and holidays
midterm exam	midterm exam	0	none			Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 8: Perfectly Competitive Markets	Perfectly competitive market	4	Combined with China's reality, such as the development and current situation of China's textile industry.	Key points: Basic characteristics of a perfectly competitive market Difficulty: Pareto efficiency		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 9: Imperfect Competition and Monopoly	Imperfect competition and monopoly	4	Combined with China's realities, such as the automotive industry and oil companies.	Key points: Characteristics of imperfectly competitive markets, the concept of monopoly Difficulty: Social welfare losses of monopoly		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 10: Competition	Competition	4	Combined with China's reality, such as the	Key points: Characteristics of oligopoly competition, Nash		Adjust the course content and progress due to the actual

among the Few	among a few		telecommunications industry.	equilibrium Difficulty: Social welfare analysis		situation of military training and holidays
Chapter 11: Economic Uncertainty	Economic uncertainty	2	Combined with China's reality, combined with the development and current situation of China's stock market.	Key points: The concepts of risk and uncertainty Difficulty: The intuitive concept and mathematical expression of deterministic equivalence		Adjust the course content and progress due to the actual situation of military training and holidays
Exercises: Chapters 8-11	Explanation of Exercises 8-11	2	Practice questions are written using Chinese cases in light of China's reality.			Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 12: Income Determination	Income determination	2	Practice questions are written using Chinese cases in light of China's reality.	Focus: Income, wealth, income distribution theory Difficulty: Factor price determination		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 13: Labor Market	labor market	2	Combined with China's reality, China's labor market and the goal of common prosperity.	Key Points: Labor supply and demand, causes of wage differentials Difficulty: Statistical discrimination		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 14: Land, Natural Resources, and the Environment	Land, natural resources and the environment	2	Integrate with China's reality and use cases and realities of China's efforts in land utilization, natural resources and environmental protection.	Focus: Land and rent, externalities Difficulty: Efficiency loss caused by externalities		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 15: Interest and Profit on Capital	Capital interest and profits	2	Based on China's reality, this article introduces the interest rates and profits paid by Chinese companies.	Key points: concepts of capital, interest, and profit		Adjust the course content and progress due to the actual situation of military training and holidays
Chapter 16: Government Taxation and Spending	Government taxation and spending	2	Based on China's reality, briefly introduce China's tax system and government spending.	Focus: Taxation, government spending		Adjust the course content and progress due to the actual situation of military training and holidays

Chapter 17: Efficiency and Equity: The Big Trade-offs	Efficiency and fairness: a major trade-off	2	In light of China's reality, we will have an in-depth discussion on the socialist market economy with Chinese characteristics and the goal of common prosperity.	Focus: The relationship between efficiency and fairness		Adjust the course content and progress due to the actual situation of military training and holidays
Exercises: Chapters 12-17	Explanation of exercises chapters 12-17	in 2	Practice questions are written using Chinese cases in light of China's reality.			Adjust the course content and progress due to the actual situation of military training and holidays
Final Exam	Final Exam	0	none			

Total: 54

(2) Teaching arrangements

(Necessary explanations should be given for the arrangement of various teaching links, such as experiments, internships, exercises, homework, etc., as well as the connection and division of labor between this course and other related courses. The arrangement of teaching links should reflect high-level, innovative and challenging characteristics.)

Classroom teaching is the main method, supplemented by classroom discussions. Homework and exercise classes are arranged, and teaching assistants correct homework.

(3) Teaching methods

(This includes classroom lectures, discussion sessions, after-class exercises and Q&A sessions. It is important to increase team learning, group assignments, the integration of experimental and theoretical classes, the use of information technology, and the shift from a teacher-centered and knowledge-centered approach to a student-centered and learning-centered approach.)

1) Lecture; 2) Homework; 3) Class discussion

(IV) Course Textbooks (The main teaching materials should be those from the "Marxism Project" and national planning textbooks as much as possible. Among similar textbooks, national planning textbooks are preferred. If there are key textbooks from the "Marxism Project" in the textbook selection range, key textbooks from the project must be used.)

Western Economics, Western Economics Editorial Group, Higher Education Press/People's Publishing House, Second Edition, 2019

Samuelson, Nordhaus: Economics, (Nineteenth Edition), Mc GRAW-HILL, 2010.

(V) Main reference books

(Recommend several reference books and indicate the title, author, publisher, version, publication date, etc., and specify a certain amount of clear reading materials for each chapter)

Mankiw, Principles of Economics (Microeconomics), Peking University Press, 2020.

(6) Method of Assessment

Course assessment should adhere to the combination of process-oriented and result-oriented assessment, adopt multiple forms and multiple stages of assessment methods, and regular scores generally account for about 40% to 60%.

Assessment method: Examination

Score entry method: Percentage

1. Attendance and homework: 10%, 1 point will be deducted for each absence, and 1 point will be deducted for each missed homework, until all points are deducted;

2. Class test 10%;

3 midterm exams 20% closed book;

4. Final exam, 60% closed-book.

Note: The syllabus should be on A4 paper, and the text should be in size 12-point Song font.

Sun Yat-sen University

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College (Department)School of Business (School of Entrepreneurship)

Course Name: Econometrics

2025

coursesyllabus

(Date of Writing: January 2025)

1. Basic information of the course

Econometrics Econometrics					
coursecategory	Special	Course Code	BSS232	Course Offering Unit	School of Business (School of Entrepreneurship)
credit	2	Study Hours	36	Teaching grade	Sophomore
For majors/categories	Economics, Business Administration				
Course Director	Zhan Yaosong				
Prerequisite Courses					
Course Objectives	This course aims to enable students to master the basic principles and methods of econometrics, lay a foundation for further study of other professional courses in economics, and have a certain ability to solve practical problems. Through learning, students can basically meet the following requirements: (1) Master the basic theories and methods of classical econometrics; (2) Be familiar with classic literature in the field; (3) Be able to apply econometric models to conduct empirical analysis of economic phenomena. The objectives of this course can be effectively achieved through a series of lectures; through assignments and final examinations, the course objectives can be measured and evaluated.				

2. Basic content of the course (requires a certain word count and cannot be too simple)

(1) Teaching schedule

Chapter order and name (detailed to section, required)	Main teaching content (required)	Required study hours (required)	Educational elements	Key points and difficulties (optional)	Week (optional)	Remarks (optional)
Chapter 1 Basic Knowledge	Calculus and matrix algebra, distribution of random variables, properties of random variables, sample description and hypothesis testing	4	Cultivate students to master professional foundation and metrology professional knowledge, have an international perspective, and understand the forefront of development of their professional disciplines.	Key points: sample description and hypothesis testing. Difficulty: Matrix algebra.		
Chapter 2 Classical Regression Model and Extension	Basic assumptions of the classical linear regression model, statistical properties of least squares estimation in finite samples, restricted least squares estimation, structural change, model stability, testing for nonlinear constraints, choosing between non-nested models, forecasting, dummy variables, functional form 11. Setting up the analysis 12. A more general test for misspecification bias	20	Cultivate students' basic ability to master the basic estimation methods of econometric regression models and apply relevant econometric theories to analyze actual cases	Focus: Model specification and estimation, inference and prediction, and functional form. Difficulty: theoretical derivation.		
Chapter 3 Aspheric Perturbations	Heteroskedasticity testing and treatment, autocorrelation testing and treatment	4	Cultivate students' understanding of basic econometric models and ability to use quantitative analysis tools, and form a thinking mode for quantitative analysis of financial problems.	Key points: LM test, DW test, cluster robust standard error, White test, GLS. Difficulty: Application of GLS method.		
Chapter 4 Panel Data Model	Panel data model setting, fixed effect model, random effect model, Hausman test	4	Ability to use professional databases to write papers and reports, and the ability to analyze quantitative results of real cases.	Key points: Fixed-effect and random-effect models. Difficulty: Hausman test method.		
Chapter 5	Basic concepts, instrumental variable		Cultivate students' ability to collect, analyze, and select relevant regression	Key points: Instrumental variable method, GMM estimation		

Endogeneity Problem	method, inference	causal	4	models, the library, cultivate professional sensitivity.	expand the method and squares method, the concept of causal inference	Difficulty: Two-stage least squares method.		
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Total: 36

(2) Teaching arrangements

(Necessary explanations should be given for the arrangement of various teaching links, such as experiments, internships, exercises, homework, etc., as well as the connection and division of labor between this course and other related courses. The arrangement of teaching links should reflect high-level, innovative and challenging characteristics.)

This course is mainly taught in the classroom, with corresponding exercises arranged for key chapters.

(3) Teaching methods

(This includes classroom lectures, discussion sessions, after-class exercises and Q&A sessions. It is important to increase team learning, group assignments, the integration of experimental and theoretical classes, the use of information technology, and the shift from a teacher-centered and knowledge-centered approach to a student-centered and learning-centered approach.)

This course integrates theory and practice, employing a teaching method that combines lectures, discussions, and student presentations. Lectures are primarily teacher-led, focusing on key questions to guide students' active thinking. Lectures fully utilize multimedia equipment, supplemented by blackboard writing, to enhance the effectiveness of the teaching process. Homework is assigned after class, which students are expected to complete individually, though group discussions are permitted. At the end of the semester, group presentations are organized to assess students' mastery and application of the foundational knowledge.

(IV) Course Textbooks (The main teaching materials should be those from the "Marxism Project" and national planning textbooks as much as possible. Among similar textbooks, national planning textbooks are preferred. If there are key textbooks from the "Marxism Project" in the textbook selection range, key textbooks from the project must be used.)

Self-compiled lecture notes

(V) Main reference books

(Recommend several reference books and indicate the title, author, publisher, version, publication date, etc., and specify a certain amount of clear reading materials for each chapter)

Stock, James H. and Mark W. Watson, "Introductions to Econometrics". Brief Edition, Pearson Addison-Wesley, Boston/Packaged with Study Guide.

(6) Method of Assessment

Course assessment should adhere to the combination of process-oriented and result-oriented assessment, adopt multiple forms and multiple stages of assessment methods, and regular scores generally account for about 40% to 60%.

Assessment method: Examination

Score entry method: Percentage

1. Composition of the total score: 50% for the final examination and 50% for regular examinations.

2. Final Examination Instructions:

The final exam is a partially open-book written exam. Students are allowed to bring one A4-sized handwritten note. The exam is graded on a 100-point scale, with the score calculated by multiplying the score by 0.5. The exam covers all chapters, with scores for each chapter being roughly evenly distributed. Questions include both calculations and essays.

III. Regular Assessment Instructions:

(1) Class participation and attendance (accounting for 10% of the total grade). 10 points will be awarded for perfect attendance; 3 points will be deducted for one absence, lateness, or early departure without reason; 6 points will be deducted for two absences; and 10 points will be deducted for three or more absences. Leave must be requested in accordance with school regulations and must be requested in advance. Leave cannot be requested on behalf of others. If leave is not requested in advance due to special circumstances, the student must complete the leave application within one week and attach an explanation of the circumstances. Otherwise, it will be treated as an absence.

(2) Classroom presentation (40% of the total score). One class presentation will be arranged according to the teaching schedule, and the content of the presentation should be a journal article designated by the course.

Note: The syllabus should be on A4 paper, and the text should be in size 12-point Song font.

Sun Yat-sen University

undergraduatecoursesyllabus

College (Department)Lingnan College

Course Name Macroeconomics

2025

coursesyllabus

(Date of Writing: January 2025)

1. Basic information of the course

Macroeconomics Macroeconomics					
coursecategory	Special	Course Code	LN134	Course Offering Unit	Lingnan College
credit	3	Study Hours	54	Teaching grade	Freshman
For majors/categories	Economics				
Course Director	Xue Chang				
Prerequisite Courses					
Course Objectives	This course is one of the principles of economics courses. The teaching goal is to enable students to have a deep understanding of the core theories and analytical logic of contemporary macroeconomics. Based on the principle of cultivating morality and educating people, students will understand and tell the story of China's macroeconomy well, enhance the "four confidences", and guided by the principle of learning for application, enable students to learn and use macroeconomic theoretical tools well, and analyze practical problems, especially the practical problems of China's macroeconomy in the new era. The basic requirements for teaching this course are the following four points: a firm grasp of the basic concepts and theories of macroeconomics; a general grasp of the ability to use macroeconomic theories to analyze real-world macroeconomic problems; a general understanding of the development of macroeconomics; and a general understanding of the macroeconomic operations of China and other major economies.				

2. Basic content of the course (requires a certain word count and cannot be too simple)

(1) Teaching schedule

Chapter order and name (detailed to section, required)	Main teaching content (required)	Required study hours (required)	Educational elements	Key points and difficulties (optional)	Week (optional)	Remarks (optional)
9.1 Gross Domestic Product and Its Measurement	Gross Domestic Product. Introducing its basic indicators that reflect macroeconomic conditions.	3	China's GDP statistics and their long-term trends help us understand the state of China's macroeconomic operations		Week 1	
10.1-10.3 Income-Expenditure Model	Determination of national income: income-expenditure model. Introduces the general principles of national income determination and the determinants of short-term national income.	3	The determination of China's national income helps us understand how China's macroeconomic operation and overall output level are determined and changed.		Week 2	
10.3, 10.4, 10.6 Income-Expenditure Model	Determination of national income: The income-expenditure model. Introducing the determinants of short-term national income and the "multiplier," an important mechanism that influences demand.	3	The determination of China's national income helps us understand how China's macroeconomic operation and overall output level are determined and changed.		Week 3	
11.1-11.2 IS-LM Model	Determination of national income: IS-LM model. Introduction to the IS curve and LM curve theory	3	Explain China's macroeconomic fluctuations and emphasize the importance of aggregate demand in balancing national income.		Week 4	
11.2-11.3 IS-LM Model	Determination of national income: IS-LM model. Introduction to the IS-LM model	3	Explain China's macroeconomic fluctuations and emphasize the importance of aggregate demand in balancing national income.		Week 5	
12.1-12.2 AD-AS Model	Determination of national income: AD-AS model. Introduction and explanation of AD curve and AS curve	3	Explain China's macroeconomic fluctuations and emphasize that the balance between China's total demand and total supply is a two-pronged approach.		Week 6	
12.2-12.3 AD-AS Model	Determination of national income: AD-AS model. Introduction to AS curve and AD-AS model	3	Explain China's macroeconomic fluctuations and emphasize that the balance between China's total demand and total supply is a two-pronged approach.		Week 7	

12.3-12.4 Model	AD-AS	Determination of national income: AD-AS model. Introduction to the AD-AS model and responses to external shocks.	3	Explain China's macroeconomic fluctuations and emphasize that the balance between aggregate demand and aggregate supply is a two-pronged approach.		Week 8	
9.2, 9.3 Price Level, Unemployment, and Their Measurement		Unemployment + Inflation. Introduce the price level and its measurement, unemployment and its measurement	3	Chinese data helps us understand two other indicators that describe or measure the overall status of an economy.		Week 9	
Midterm quiz		Midterm quiz	3	Midterm quiz		Week 10	
13.1, Unemployment and Inflation	13.2 and	Unemployment + Inflation. Introducing the macroeconomic explanation of unemployment and inflation	3	Chinese data helps us understand the cyclical fluctuations in China's economic development.		Week 11	
13.3-13.4 The Phillips Curve and the Business Cycle		Phillips Curve + Business Cycle. Introduce and explain the Phillips Curve, Expectations Theory, and Business Cycle Theory.	3	China's economic cycle, enlightening understanding of economic cycle phenomena such as unemployment and inflation under China's market economy		Week 12	
10.5, 14.1, 14.2 Macroeconomics under Open Conditions		Open Macroeconomics. Introduction to foreign demand, balance of payments, and exchange rates.	3	Chinese data helps us understand some facts about China's open economy		Week 13	
14.2-14.4 Macroeconomics under Open Conditions		Open Macroeconomics. Introducing the balance of payments and exchange rates, balance of payments adjustments under a fixed exchange rate system, and balance of payments adjustments under a floating exchange rate system.	3	Explain the characteristics of China's open economy and learn to analyze macroeconomic policy issues under China's open economy		Week 14	
16.1-16.3 Economic Growth	Economic	Economic growth. Introduce and explain the description and facts of economic growth, the neoclassical growth model, and policies to promote economic growth.	3	Explain China's economic growth and help understand some facts and great achievements of China's economic growth		Week 15	
16.3-16.6 Economic Growth	Economic	Economic growth. Introduce and explain the description and facts of economic growth, the neoclassical growth model, the endogenous growth model, and policies to promote economic growth.	3	Explain China's economic growth and help understand the determinants and great achievements of China's economic growth		Week 16	
				China's macroeconomic control policies and			

15.1-15.3 Macroeconomic Policy Management	Macroeconomic policy management. Introduce and explain macroeconomic policy objectives, fiscal policy, and monetary policy.	3	successful experiences help us understand how China can effectively coordinate the use of monetary policy tools and fiscal policy tools		Week 17	
15.4-15.5 Macroeconomic Policy Management	Macroeconomic policy management. Introduce and explain the limitations and coordination of fiscal and monetary policies.	3	China's macroeconomic control policies and successful experiences help us understand how China can effectively coordinate the use of monetary policy tools and fiscal policy tools		Week 18	

Total: 54

(2) Teaching arrangements

(Necessary explanations should be given for the arrangement of various teaching links, such as experiments, internships, exercises, homework, etc., as well as the connection and division of labor between this course and other related courses. The arrangement of teaching links should reflect high-level, innovative and challenging characteristics.)

There will be random attendance checks in class, two homework assignments (primarily textbook exercises, one midterm and one final assignment, tentatively due in Weeks 9 and 18), one quiz in Week 10, and one closed-book final exam. Classes will be postponed during holidays, but the Week 10 quiz will not be postponed.

(3) Teaching methods

(This includes classroom lectures, discussion sessions, after-class exercises and Q&A sessions. It is important to increase team learning, group assignments, the integration of experimental and theoretical classes, the use of information technology, and the shift from a teacher-centered and knowledge-centered approach to a student-centered and learning-centered approach.)

Classroom Lectures

After-class exercises and Q&A

(IV) Course Textbooks (The main teaching materials should be those from the "Marxism Project" and national planning textbooks as much as possible. Among similar textbooks, national planning textbooks are preferred. If there are key textbooks from the "Marxism Project" in the textbook selection range, key textbooks from the project must be used.)

Western Economics (Volume 2, Second Edition), Western Economics Editorial Group, Higher Education Press

(V) Main reference books

(Recommend several reference books and indicate the title, author, publisher, version, publication date, etc., and specify a certain amount of clear reading materials for each chapter)

Xi Jinping: The Governance of China

Guidance Notes on the 19th National Congress of the Communist Party of China

Guidance Notes on the Report to the 20th National Congress of the Communist Party of China

State Council Executive Meeting http://www.gov.cn/guowuyuan/gwy_cwh.htm

National Bureau of Statistics of China <http://www.stats.gov.cn/>

(6) Method of Assessment

Course assessment should adhere to the combination of process-oriented and result-oriented assessment, adopt multiple forms and multiple stages of assessment methods, and regular scores generally account for about 40% to 60%.

Assessment method: Examination

Score entry method: Percentage

Regular grades (5% for attendance, 5% for homework, 30% for midterm quizzes), totaling 40%

Final grade (60% of the final closed-book exam), total 60%

Note: The syllabus should be on A4 paper, and the text should be in size 12-point Song font.

Sun Yat-sen University

undergraduatecoursesyllabus

College (Department)School of Mathematics

Course Name: Advanced Mathematics I (II)

2024

coursesyllabus

(Date of writing: December 2024)

1. Basic information of the course

Advanced Mathematics I (II) Advanced Mathematics-1(II)					
coursecategory	Special	Course Code	MA190	Course Offering Unit	School of Mathematics
credit	5	Study Hours	90	Teaching grade	First-year undergraduate
For majors/categories	Optoelectronics Science, Microelectronics, Electronic Information Science and Technology				
Course Director	Huang Jingchi				
Prerequisite Courses	Advanced Mathematics I				
Course Objectives	This course mainly provides the necessary basic mathematical knowledge for subsequent mathematics courses and professional courses, improves students' mathematical literacy, and cultivates students' abstract thinking ability, problem summarization ability, logical reasoning ability, spatial imagination ability and self-study ability, as well as students' calculation ability and the ability to use the knowledge learned to analyze and solve practical problems.				

2. Basic content of the course (requires a certain word count and cannot be too simple)

(1) Teaching schedule

Chapter order and name (detailed to section, required)	Main teaching content (required)	Required study hours (required)	Educational elements	Key points and difficulties (optional)	Week (optional)	Remarks (optional)
	Teaching requirements:					

\$7.1 The concept and properties of double integrals \$7.2 Calculation of double integrals	Understand the concept of multiple integrals, master the calculation method of double integrals (rectangular coordinates, polar coordinates) and master the calculation method of triple integrals (rectangular coordinates, cylindrical coordinates, spherical coordinates) Key points: the concept of multiple integrals and the calculation method of multiple integrals. Difficulty: How to convert double integrals and triple integrals into cumulative integrals.	6	The concept of multiple integral is the extension of the concept of definite integral of a univariate function to the case of a multivariate function. It has a wide range of practical backgrounds and application values.	Key points: the concept of multiple integrals and the calculation method of multiple integrals. Difficulty: How to convert double integrals and triple integrals into cumulative integrals.	Week 1	
\$7.3 The concept and calculation of triple integrals \$7.4 Examples of geometric applications of multiple integrals	Teaching requirements: Understand the concept of multiple integrals, master the calculation method of double integrals (rectangular coordinates, polar coordinates) and master the calculation method of triple integrals (rectangular coordinates, cylindrical coordinates, spherical coordinates) Key points: the concept of multiple integrals and the calculation method of multiple integrals. Difficulty: How to convert double integrals and triple integrals into cumulative integrals.	6	The concept and calculation of triple integrals are needed when calculating the mass, center of mass, and other problems of space objects with uneven density distribution.	Key points: the concept of multiple integrals, the calculation method of multiple integrals and the geometric application of multiple integrals. Difficulties: Methods of converting double integrals and triple integrals into cumulative integrals and geometric applications of multiple integrals.	Week 2	
	Teaching Requirements: Understand the concepts of two types of curve integrals and be able to transform them into definite integrals using parametric equations of curves. Understand the concepts of two types of surface integrals and be able to calculate them. Understand and master Green's formula and be able to simplify the calculation of integrals		The curve integrals	Key Points: Concepts and calculation methods of two types		

§8.1 Curvilinear integrals of the first type §8.2 Curvilinear integrals of the second type	using the path-independent condition for plane curve integrals. Master Gauss's formula and understand Stokes's formula. Understand the concepts of gradient, divergence, and curl. Key Points: Concepts and calculation methods of two types of curve integrals and two types of surface integrals. Green's formula, Gauss's formula, and Stokes' formula. Conditions for curve integrals to be path-independent. Difficulties: Calculation of curve integrals and surface integrals, Green's formula, Gauss's formula.	6	and curved areas in this chapter also have a rich practical background. According to the different requirements of physics or geometry, they are conducive to the cultivation of basic theories of different majors.	of curve integrals and two types of surface integrals. Green's formula, Gauss's formula, and Stokes' formula. Conditions for curve integrals to be path-independent. Difficulties: Calculation of curve integrals and surface integrals, Green's formula, Gauss's formula.	Week 3	
§8.3 Green's formula, plane curve integral of the second type and path-independent conditions §8.4 Surface integrals of the first type	Teaching Requirements: Understand the concepts of two types of curve integrals and be able to transform them into definite integrals using parametric equations of curves. Understand the concepts of two types of surface integrals and be able to calculate them. Understand and master Green's formula and be able to simplify the calculation of integrals using the path-independent condition for plane curve integrals. Master Gauss's formula and understand Stokes's formula. Understand the concepts of gradient, divergence, and curl. Key Points: Concepts and calculation methods of two types of curve integrals and two types of surface integrals. Green's formula, Gauss's formula, and Stokes' formula. Conditions for curve integrals to be path-independent. Difficulties: Calculation of curve integrals and surface	6	From its mathematical form, Green's formula is equivalent to the generalization of the fundamental theorem of calculus in a certain sense. In nature, it is a reflection of many physical phenomena and has become the basis for the study of many plane physical fields.	Key Points: Concepts and calculation methods of two types of curve integrals and two types of surface integrals. Green's formula, Gauss's formula, and Stokes' formula. Conditions for curve integrals to be path-independent. Difficulties: Calculation of curve integrals and surface integrals, Green's formula, Gauss's formula.	Week 4	

	integrals, Green's formula, Gauss's formula.					
§8.5 Surface integrals of the second type §8.6 Gauss formula and Stokes formula §8.7 Introduction to Field Theory	Teaching Requirements: Understand the concepts of two types of curve integrals and be able to transform them into definite integrals using parametric equations of curves. Understand the concepts of two types of surface integrals and be able to calculate them. Understand and master Green's formula and be able to simplify the calculation of integrals using the path-independent condition for plane curve integrals. Master Gauss's formula and understand Stokes's formula. Understand the concepts of gradient, divergence, and curl. Key Points: Concepts and calculation methods of two types of curve integrals and two types of surface integrals. Green's formula, Gauss's formula, and Stokes' formula. Conditions for curve integrals to be path-independent. Difficulties: Calculation of curve integrals and surface integrals, Green's formula, Gauss's formula.	6	In a certain sense, Gauss's formula and Stokes' formula are generalizations of Green's formula. The three are interrelated and have different application scopes in different spaces and fields.	Key Points: Concepts and calculation methods of two types of curve integrals and two types of surface integrals. Green's formula, Gauss's formula, and Stokes' formula. Conditions for curve integrals to be path-independent. Difficulties: Calculation of curve integrals and surface integrals, Green's formula, Gauss's formula.	Week 5	
	Teaching Requirements: Understand concepts such as differential equations, solutions, general solutions, initial conditions, and particular solutions. Master elementary integration methods for solving first-order differential equations. Understand the structure of general solutions to linear differential equations and master the methods for solving second-order linear homogeneous equations with			Key points: Separable equations, homogeneous equations, first-order		

§9.1 Basic Concepts §9.2 Elementary Integration	constant coefficients. Master the methods for solving second-order linear non-homogeneous equations with constant coefficients when the non-homogeneous terms are in the form of polynomials, exponential functions, sine functions, and cosine functions. Key points: Separable equations, homogeneous equations, first-order linear differential equations, Bernoulli equations, total differential equations. Constant variation method. The structure of solutions to linear differential equations, and solutions to second-order linear homogeneous and nonhomogeneous equations with constant coefficients. Difficulty: Reduced-order solutions to some high-order equations. Structure and solution of general solutions to linear differential equations.	6	We hope to have a point of view that unifies these formulas into one formula. This approach will not only make it easier to remember these formulas, but more importantly, it will make it easier to extend these formulas to high-dimensional spaces.	linear differential equations, Bernoulli equations, total differential equations. Constant variation method. The structure of solutions to linear differential equations, and solutions to second-order linear homogeneous and nonhomogeneous equations with constant coefficients. Difficulty: Reduced-order solutions to some high-order equations. Structure and solution of general solutions to linear differential equations.	Week 6
§9.3 Existence and uniqueness theorems of solutions to differential equations	Teaching Requirements: Understand concepts such as differential equations, solutions, general solutions, initial conditions, and particular solutions. Master elementary integration methods for solving first-order differential equations. Understand the structure of general solutions to linear differential equations and master the methods for solving second-order linear homogeneous equations with constant coefficients. Master the methods for solving second-order linear non-homogeneous equations with constant coefficients when the non-homogeneous terms are in the form of	6	The theory of ordinary differential equations has a long history of development. It was almost created at the same time as calculus and has always been an important branch of mathematics that	Key points: Separable equations, homogeneous equations, first-order linear differential equations, Bernoulli equations, total differential equations. Constant variation method. The structure of solutions to linear differential equations, and solutions to second-order linear	Week 7

§9.4 Higher-order linear differential equations	polynomials, exponential functions, sine functions, and cosine functions. Key points: Separable equations, homogeneous equations, first-order linear differential equations, Bernoulli equations, total differential equations. Constant variation method. The structure of solutions to linear differential equations, and solutions to second-order linear homogeneous and nonhomogeneous equations with constant coefficients. Difficulty: Reduced-order solutions to some high-order equations. Structure and solution of general solutions to linear differential equations.		combines practice. In natural sciences and engineering the mathematical models of many problems are ordinary differential equations.	homogeneous and nonhomogeneous equations with constant coefficients. Difficulty: Reduced-order solutions to some high-order equations. Structure and solution of general solutions to linear differential equations.		
§9.5 Second-order linear differential equations with constant coefficients §9.6 Constant Variation Method and Euler Equation	Teaching Requirements: Understand concepts such as differential equations, solutions, general solutions, initial conditions, and particular solutions. Master elementary integration methods for solving first-order differential equations. Understand the structure of general solutions to linear differential equations and master the methods for solving second-order linear homogeneous equations with constant coefficients. Master the methods for solving second-order linear non-homogeneous equations with constant coefficients when the non-homogeneous terms are in the form of polynomials, exponential functions, sine functions, and cosine functions. Key points: Separable equations, homogeneous equations, first-order linear differential	6	Ordinary differential equations are important tools for studying certain scientific and technological problems, especially many mechanical and physical problems.	Key points: Separable equations, homogeneous equations, first-order linear differential equations, Bernoulli equations, total differential equations. Constant variation method. The structure of solutions to linear differential equations, and solutions to second-order linear homogeneous and nonhomogeneous equations with constant coefficients. Difficulty: Reduced-order solutions to some high-order equations. Structure and solution of general	Week 8	

	equations, Bernoulli equations, total differential equations. Constant variation method. The structure of solutions to linear differential equations, and solutions to second-order linear homogeneous and nonhomogeneous equations with constant coefficients. Difficulty: Reduced-order solutions to some high-order equations. Structure and solution of general solutions to linear differential equations.			solutions to linear differential equations.		
§10.1 Cauchy's Convergence Principle and the Concept of Numerical Series §10.2 Convergence Criteria for Positive Series	Teaching Requirements: Understand the concepts of convergence and divergence of series. Understand the necessary conditions for series convergence. Master the ratio and root value convergence methods for positive series. Be familiar with the convergence and divergence of geometric series and P-series. Master the absolute and conditional convergence of arbitrary series. Understand the concept of uniform convergence of function series and how to determine it. Master the convergence region of power series and the method for finding sum functions, and understand the basic properties of power series within their convergence intervals. Master the method of expanding functions into power series. Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions.	6	The concept and theory of infinite series is a crucial and significant component of analysis. People have long used infinite series to represent functions, particularly solutions to differential equations. Infinite series, along with calculus and the calculus of variations, formed the backbone of analysis in the 17th and 18th centuries.	Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions. Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.	Week 9	

		Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.				
§10.3 Series	Arbitrary	Teaching Requirements: Understand the concepts of convergence and divergence of series. Understand the necessary conditions for series convergence. Master the ratio and root value convergence methods for positive series. Be familiar with the convergence and divergence of geometric series and P-series. Master the absolute and conditional convergence of arbitrary series. Understand the concept of uniform convergence of function series and how to determine it. Master the convergence region of power series and the method for finding sum functions, and understand the basic properties of power series within their convergence intervals. Master the method of expanding functions into power series. Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions. Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.	4	The emergence of series stems from practical production, and they also have a vast application space in practice. Series are used in mathematics and other fields, such as using series theory to transform complex problems such as finding limits and approximate calculations into simple series problems.	Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions.	Week 10

		and expanding functions into power series.				
\$10.4 Series	Functional	Teaching Requirements: Understand the concepts of convergence and divergence of series. Understand the necessary conditions for series convergence. Master the ratio and root value convergence methods for positive series. Be familiar with the convergence and divergence of geometric series and P-series. Master the absolute and conditional convergence of arbitrary series. Understand the concept of uniform convergence of function series and how to determine it. Master the convergence region of power series and the method for finding sum functions, and understand the basic properties of power series within their convergence intervals. Master the method of expanding functions into power series. Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions. Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.	4	Series is an important tool for studying functions, and plays an important role both in theory and in practical applications. This is because: on the one hand, series can be used to represent many commonly used non-elementary functions, and the solutions to differential equations are often represented by series; on the other hand, functions can be expressed as series, and thus used to study functions, such as using power series to study non-elementary functions and perform approximate calculations.	Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions. Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.	Week 11
		Teaching Requirements: Understand the concepts of				

§10.5 Power Series §10.6 Taylor series	convergence and divergence of series. Understand the necessary conditions for series convergence. Master the ratio and root value convergence methods for positive series. Be familiar with the convergence and divergence of geometric series and P-series. Master the absolute and conditional convergence of arbitrary series. Understand the concept of uniform convergence of function series and how to determine it. Master the convergence region of power series and the method for finding sum functions, and understand the basic properties of power series within their convergence intervals. Master the method of expanding functions into power series. Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions. Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.	4	The power series can be regarded as an "infinite degree polynomial", which is a powerful tool in both theoretical application and practical verification.	Key points: Methods for determining the convergence or divergence of numerical series, methods for finding the sum function of power series, and power series expansions of common functions. Difficulties: Methods for determining convergence and divergence of series, concepts of absolute convergence, conditional convergence and uniform convergence, and expanding functions into power series.	Week 12
	Teaching Requirements: Understand the concepts of convergence and divergence of generalized integrals. Master the methods for determining the convergence and divergence of generalized integrals. Master the methods for			Key points: Methods	

§11.1 Generalized integrals	determining the absolute and conditional convergence of generalized integrals, including the properties and calculations of normal and generalized integrals with parameters. Key points: Methods for determining convergence and divergence of generalized integrals. Absolute convergence, conditional convergence, and their determination methods; properties and calculations of normal integrals and generalized integrals with parameters. Difficulties: Absolute convergence, conditional convergence and its judgment methods, calculation of normal integrals and generalized integrals with parameters.	4	The generalized integral is a generalization of the concept of ordinary definite integral. Many practical problems require consideration of the generalized integral on an infinite interval.	for determining convergence and divergence of generalized integrals. Absolute convergence, conditional convergence, and their determination methods; properties and calculations of normal integrals and generalized integrals with parameters. Difficulties: Absolute convergence, conditional convergence and its judgment methods, calculation of normal integrals and generalized integrals with parameters.	Week 13	
§11.2 Normal integrals with parameters §11.3 Generalized integrals with parameters	Teaching Requirements: Understand the concepts of convergence and divergence of generalized integrals. Master the methods for determining the convergence and divergence of generalized integrals. Master the methods for determining the absolute and conditional convergence of generalized integrals, including the properties and calculations of normal and generalized integrals with parameters. Key points: Methods for determining convergence and divergence of generalized integrals. Absolute convergence, conditional convergence, and their determination methods; properties and calculations of	4	The practical value of Theorem 5 is that it provides a method for finding the value of certain infinite integrals.	Key points: Methods for determining convergence and divergence of generalized integrals. Absolute convergence, conditional convergence, and their determination methods; properties and calculations of normal integrals and generalized integrals with parameters. Difficulties: Absolute convergence, conditional convergence and its judgment methods, calculation of normal integrals and generalized integrals with parameters.	Week 14	

	normal integrals and generalized integrals with parameters. Difficulties: Absolute convergence, conditional convergence and its judgment methods, calculation of normal integrals and generalized integrals with parameters.					
§12.1 Trigonometric Function Systems and Their Orthogonality	Teaching Requirements: Understand the significance and necessity of trigonometric series expansion, as well as the conditions and methods for expanding functions into trigonometric series, and understand the significance of Bessel's inequality and Paschal's equation.	4	Generalized integral is a modern mathematics that has developed along with mathematics. As a basic mathematical proposition, it is an important concept in higher mathematics. Its emergence has solved many computational problems in physics and has also promoted the development of other sciences.	Key points: Methods for determining convergence and divergence of generalized integrals. Absolute convergence, conditional convergence and their determination, properties and calculations of normal integrals and generalized integrals with parameters. Difficulties: Absolute convergence, conditional convergence and how to determine it, and the calculation of normal and generalized integrals involving parameters. Key Difficulties: Conditions for and methods of expanding trigonometric series.	Week 15	
§12.2 Fourier series of periodic functions and their convergence	Teaching Requirements: Understand the significance and necessity of trigonometric series expansion, as well as the conditions and methods for expanding functions into trigonometric series, and understand the significance of Bessel's inequality and Paschal's equation.	4	This chapter discusses the problem of expanding a periodic function into a trigonometric series. This problem is not only of great value in mathematical theory research, but also has wide applications in other disciplines and engineering technologies.	Key points and difficulties: conditions and methods for expanding trigonometric series.	Week 16	
§12.3 Bessel's inequality and Pasval's equation Summary Review	Teaching Requirements: Understand the significance and necessity of trigonometric series expansion, as well as the conditions and methods for expanding functions into trigonometric series, and understand the significance of Bessel's inequality and Paschal's equation.	4	The two formulas, Bessel's inequality and Pasval's equation, have learning and application value in many aspects, especially in studying mechanics,	Key points and difficulties: conditions and methods for expanding	Week 17	

	methods for expanding functions into trigonometric series, and understand the significance of Bessel's inequality and Pascal's equation.		architecture, engineering, computers, etc.	trigonometric series.		
Final Exam	none	4	The classroom is closely linked to the fundamental task of cultivating morality and educating people, with the socialist core values and the education of China's excellent traditional culture as the soul and main line, with a deep sense of family and country as the mind, and with mathematical knowledge teaching as the carrier. It does a good job in all-round talent training such as classroom education, textbook education, practical education, and scientific research education, and subtly improves students' ideological and moral qualities and political literacy in the process of mathematical knowledge teaching, so that students can strengthen the "four self-confidences" and achieve "two safeguards" while mastering mathematical knowledge, and grow into qualified builders and successors with all-round development of morality, intelligence, physical fitness, aesthetics, and labor.	none	Week 18	
Final Exam	none	0	none	none	Week 19	none

Total: 90

(2) Teaching arrangements

(Necessary explanations should be given for the arrangement of various teaching links, such as experiments, internships, exercises, homework, etc., as well as the connection and division of labor between this course and other related courses. The arrangement of teaching links should reflect high-level, innovative and challenging characteristics.)

Homework is assigned every week. Since class time is tight and it is difficult to have special class time for exercises, we try to insert the explanation of homework exercises into the class time as much as possible.

(3) Teaching methods

(This includes classroom lectures, discussion sessions, after-class exercises and Q&A sessions. It is important to increase team learning, group assignments, the integration of experimental and theoretical classes, the use of information technology, and the shift from a teacher-centered and knowledge-centered approach to a student-centered and learning-centered approach.)

The main teaching methods are topic lectures and topic questions. Questions are answered promptly during and after class, and homework is

assigned every week.

(IV) Course Textbooks (The main teaching materials should be those from the "Marxism Project" and national planning textbooks as much as possible. Among similar textbooks, national planning textbooks are preferred. If there are key textbooks from the "Marxism Project" in the textbook selection range, key textbooks from the project must be used.)

"Advanced Mathematics" (3rd edition), Volume 2, edited by Li Zhong et al., Peking University Press, June 1, 2023.

(V) Main reference books

(Recommend several reference books and indicate the title, author, publisher, version, publication date, etc., and specify a certain amount of clear reading materials for each chapter)

(1) A Concise Course in Mathematical Analysis (2nd Edition), Deng Donggao and Yin Xiaoling, Higher Education Press, April 1, 2010

(2) Mathematical Analysis (4th Edition), Volumes 1 and 2, compiled by the Department of Mathematics, East China Normal University, Higher Education Press, July 1, 2010

(3) Advanced Mathematics (6th Edition), Volumes 1 and 2, Department of Mathematics, Tongji University, Higher Education Press, April 1, 2007

(6) Method of Assessment

Course assessment should adhere to the combination of process-oriented and result-oriented assessment, adopt multiple forms and multiple stages of assessment methods, and regular scores generally account for about 40% to 60%.

Assessment method: Examination

Score entry method: Percentage

Overall score:

Regular grades account for 40%, of which 1. Regular homework accounts for 20%, with about 10-20 homeworks, each accounting for about 1%-2%; 2. Class quizzes account for 20%, with 4-10 quizzes, each accounting for 2%-5%.

The final exam accounts for 60% of the score (closed-book written exam).

Note: The syllabus should be on A4 paper, and the text should be in size 12-point Song font.

Sun Yat-sen University

undergraduatecoursesyllabus

College (Department)School of Mathematics (Zhuhai)

Course Name: Probability and Statistics (Business Management)

2025

coursesyllabus

(Date of Writing: July 2025)

1. Basic information of the course

Probability and Statistics (Business Management) Probability Theory and Statistics.					
coursecategory	Special	Course Code	MAZ117	Course Offering Unit	School of Mathematics (Zhuhai)
credit	3	Study Hours	54	Teaching grade	Sophomore
For majors/categories	Tourism Management				
Course Director	Kuang Dongyang				
Prerequisite Courses	Advanced Mathematics, Linear Algebra				
Course Objectives	"Probability and Statistics (Business and Management)" is a required course for sophomores, with a closed-book final exam. This course covers the basic theories and methods of probability theory and mathematical statistics. It aims to help students understand and master the fundamental concepts and methods involved in probability theory and mathematical statistics, and to cultivate their ability to analyze and solve problems using the concepts and methods of probability theory and mathematical statistics. Firmly grasp: the addition formula, multiplication formula, total probability formula, and Bayesian formula of probability; the distribution of random variables (including two-dimensional random variables); several important distributions; basic concepts of mathematical statistics; moment estimation and maximum likelihood estimation. Master: the concept of probability; probability model of independent experiments; distribution of functions of random variables; mathematical expectation and variance of random variables; covariance and correlation				

coefficient of two-dimensional random variables; sample mean and sample variance.
Understand: the concept of probability space; the law of large numbers and the central limit theorem; point estimation and interval estimation of normal populations; hypothesis testing of normal population parameters. During teaching, the philosophical ideas contained in the teaching content are deeply and effectively explored, and instilled into students during the teaching process, so that students can learn knowledge and understand the basic principles such as truth is infinite but human cognition of the world is limited, and understand the essence of probability theory and mathematical statistics courses; at the same time, the important role of this course in tourism management, meteorology and natural disaster forecasting, etc. is introduced to stimulate students' interest in learning and patriotism.

2. Basic content of the course (requires a certain word count and cannot be too simple)

(1) Teaching schedule

Chapter order and name (detailed to section, required)	Main teaching content (required)	Required study hours (required)	Educational elements	Key points and difficulties (optional)	Week (optional)	Remarks (optional)
Chapter 1 Random Events and Probability 1.1 Random Events 1.2 Probability of random events	Random events, sample spaces, and the relationship and operation of events; frequency and probability of events;	4	Students are introduced to the history of probability theory and mathematical statistics, as well as their applications in various disciplines. This stimulates their interest and enthusiasm, helps them understand what they are learning and what they will accomplish after completing their studies, and encourages them to study diligently from the perspective of a sense of social responsibility. This allows students to grasp the "scientific spirit" and strengthen their sense of social responsibility. The origins of this discipline are related to gambling, which is illegal.			Key points: Definition and properties of probability;
Chapter 1 Random Events and Probability 1.3 Classical probability models and geometric probability models	Classical probability model; axiomatic definition of probability; Conditional probability, addition formula, multiplication	4	The classical probability model is used to calculate the probability of winning the lottery and the probability of winning a gamble. Through the calculation results, students intuitively feel that winning the lottery and winning the gamble are low-probability events, which are almost			Key points: conditional probability, addition formula, total probability formula and Bayes' formula;

1.4 Conditional Probability	formula, total probability formula, Bayesian formula;		impossible to happen in an experiment in real life. Such results warn students not to expect to make money through speculation and luck, but to work hard and down-to-earth.			independence of events;
Chapter 1 Random Events and Probability 1.5 Independence of Events	Definition of Independent Distribution Chapter 1 Review	4	The total probability formula and the Bayesian formula are used to calculate the probability of passing the exam by guessing without studying hard. This is a very small probability. This result tells students not to expect to get something for nothing, but to study hard and down-to-earth to achieve good results. Whether in study or life, students must establish correct values and must have a truth-seeking and pragmatic spirit.			
Chapter 2 Distribution and Numerical Characteristics of Random Variables 2.1 Random variables and their distribution 2.2 Numerical Characteristics of Random Variables	Random variables and their distribution functions; Mathematical expectation and higher-order moments of random variables;	4	Incorporate ideological and political elements into teaching.			Key points: The concepts and properties of distribution law (density function) and distribution function;
Chapter 2 Distribution and Numerical Characteristics of Random Variables 2.3 Commonly used discrete distributions	Discrete random variables and their distribution laws, hypergeometric distribution, binomial distribution, Poisson distribution;	4	Incorporate ideological and political elements into teaching.			
Chapter 2 Distribution and Numerical Characteristics of Random Variables 2.4 Commonly used continuous distributions	Continuous random variables and their density functions, uniform distribution, exponential distribution, and normal distribution;	4	Through practical examples, students are taught not to trust online rumors and not to participate in online violence.			
Chapter 2 Distribution and Numerical Characteristics of Random Variables 2.5 Distribution of functions of random	The distribution of a function of a random variable; the mathematical expectation of a function of a random variable.	4	Incorporate ideological and political elements into teaching.			

variables	Chapter 2 Review					
Chapter 3 Random Vectors 3.1 Distribution of random vectors	Random vectors and their joint distribution function; the joint distribution law of discrete two-dimensional random vectors and the joint probability density of continuous two-dimensional random vectors;	4	Incorporate ideological and political elements into teaching.			
Chapter 3 Random Vectors 3.2 Conditional Distribution and Independence of Random Variables	Independence of random vectors; marginal and conditional distributions;	4	Incorporate ideological and political elements into teaching.			
Chapter 3 Random Vectors 3.3 Distribution and mathematical expectation of functions of random vectors	Two-dimensional uniform distribution; bivariate normal distribution; distribution of functions of two-dimensional random vectors; covariance and correlation coefficient;	2	In the teaching of probability theory and mathematical statistics, mathematical expectation and variance are used to explain the bucket law, thereby illustrating the importance of team strength and cultivating students' team spirit.			Key points: mathematical expectation and variance;
Chapter 3 Random Vectors 3.4 Numerical characteristics of random vectors;	Covariance matrix, correlation matrix, conditional expectation	2	The calculation results of actual examples intuitively illustrate the importance of integrity. Tell students that integrity is the basic requirement for us to behave in the world. Integrity is hard to come by and easy to lose, so we must abide by it.			Key points: Covariance and correlation coefficient and their properties; mathematical expectation of functions of random variables;
Chapter 3 Random Vectors 3.5 The Law of Large Numbers and the Central Limit Theorem	According to the definition of probability convergence; Mathematical Description of the Law of Large Numbers and the Central Limit	2	The law of large numbers describes the statistical pattern of experimental results when a large number of independent experiments are repeated. According to the law, the greater the number of independent replicates, the larger the sample size (in statistics), and the closer the average of the experimental results is to the mathematical expectation. The law of large			Key points: Law of large numbers and central limit theorem under independent and identically distributed. Difficulties: Chebyshev's inequality; law of

	Theorem Chapter 3 Review		numbers demonstrates that the development of things exists within a dialectical unity of opposites. By studying the law of large numbers, students can understand the dialectical relationship between quantitative change and qualitative change.			large numbers and central limit theorem.
Chapter 4 Basics of Mathematical Statistics 4.1 Population and Sample 4.2 Statistics	Population distribution and sample distribution Some common statistics based on moments of various orders and their significance	2	Based on mathematical statistics and probability theory, this study studies the statistical regularity of random phenomena. By organizing, grouping, calculating, and summarizing sample data, it makes inferences and predictions about the population.			Key points: Statistics, distribution of sample statistics, chi-square distribution, t distribution and F distribution; sampling theory of normal population.
Chapter 4 Basics of Mathematical Statistics 4.3 Commonly used statistical distributions 4.4 Sampling Distribution	Chi-square distribution Student-t distribution F-distribution Chapter 4 Review	2	Same as above			
Chapter 5 Parameter Estimation and Hypothesis Testing 5.1 Overview of Point Estimation	Definition and application scenarios of point estimation Criteria for evaluating estimators	2	Statistics is a kind of knowledge from experience to rationality. It is a science that uses randomness to discover regularity. There is inevitability in randomness. This is the philosophy of statistics.			Key points: Point estimation method; selection criteria for point estimators.
Chapter 5 Parameter Estimation and Hypothesis Testing 5.2 Maximum Likelihood Estimation and Moment Estimation of Parameters	Maximum Likelihood Estimation Moment Estimation	2	Statistics cannot give you 100% accuracy, but seeks a reasonable range of accuracy.			
Chapter 5 Parameter Estimation and Hypothesis Testing 5.3 Confidence Intervals 5.4 Overview of Hypothesis Testing	Confidence interval calculation based on normal distribution Asymptotic confidence intervals for large samples The idea, general steps and significance level of hypothesis testing	2	Statistics cannot give you 100% correctness, but it seeks a reasonable range of accuracy; statistics cannot tell you what the future will look like, but it tells a trend; statistics cannot stand alone, but it			

	Two types of errors in statistical inference		integrates other disciplines to prosper the entire earth!			
Chapter 5 Parameter Estimation and Hypothesis Testing 5.5 Parametric Hypothesis Testing on Mean and Variance for a Single Normal Population	Hypothesis Testing Chapter 5 Review	2	Same as above			

Total: 54

(2) Teaching arrangements

(Necessary explanations should be given for the arrangement of various teaching links, such as experiments, internships, exercises, homework, etc., as well as the connection and division of labor between this course and other related courses. The arrangement of teaching links should reflect high-level, innovative and challenging characteristics.)

The course is mainly based on classroom teaching, supplemented by discussion of exercises and question-answering during and after class.

A summary of the knowledge points is made in each chapter, homework is assigned every week, and exercises on relevant knowledge points are explained during each class.

(3) Teaching methods

(This includes classroom lectures, discussion sessions, after-class exercises and Q&A sessions. It is important to increase team learning, group assignments, the integration of experimental and theoretical classes, the use of information technology, and the shift from a teacher-centered and knowledge-centered approach to a student-centered and learning-centered approach.)

The main teaching method is classroom lectures, interspersed with questions and discussions. Exercises attached to the textbook are left as homework after each class every week. Students' questions about the course knowledge points are answered at any time during breaks, after class, and at other times, either face-to-face or by email.

There is a fixed question-and-answer period every week, during which teachers and students, teaching assistants and students, and students and students can organize discussions on probability theory and mathematical statistics problems.

(IV) Course Textbooks (The main teaching materials should be those from the "Marxism Project" and national planning textbooks as much as possible. Among similar textbooks, national planning textbooks are preferred. If there are key textbooks from the "Marxism Project" in the textbook selection range, key textbooks from the project must be used.)

Probability Theory and Mathematical Statistics (3rd Edition), Long Yonghong, Higher Education Press, June 2009.

(V) Main reference books

(Recommend several reference books and indicate the title, author, publisher, version, publication date, etc., and specify a certain amount of clear reading materials for each chapter)

Probability Theory and Mathematical Statistics (4th Edition), Zhejiang University, Higher Education Press, October 2010

(6) Method of Assessment

Course assessment should adhere to the combination of process-oriented and result-oriented assessment, adopt multiple forms and multiple stages of assessment methods, and regular scores generally account for about 40% to 60%.

Assessment method: Examination

Score entry method: Percentage

The exam is closed-book.

The total score is 100 points = regular scores 100*50% + final exam 100*50%.

Regular score of 100 points = midterm exam 100*40% + regular attendance 100*10% + homework 100*50%

Homework is required to be submitted once a week and will be graded A, B, C, and D. A gets 4 points, B gets 3 points, C gets 2 points, and D gets 1 point. If the homework is not submitted or plagiarism is found, the score will be 0.

Note: The syllabus should be on A4 paper, and the text should be in size 12-point Song font.

2.1 Preferences and Choices

2.2 Budget Constraints

2.3 Consumer Choice

2.4 Revealed Preference Principle

2.5 Slutsky's Law of Compensation and Demand

Consumer Utility Theory

Lesson objectives: Understand the properties of the utility function and use the utility maximization model to analyze consumer behavior.

3.1 Utility Function

3.2 Properties of the utility function

3.3 Mathematical Problems of Optimization

3.4 Utility maximization

3.5 Requirements

3.6 Slutsky equation

3.7 Application of the Slutsky equation

Consumer Welfare Measurement

Lesson Objective: Understand how economics measures consumer welfare.

4.1 Overall welfare measurement

4.2 Willingness to Pay

4.3 Equivalent Change and Compensatory Change

4.4 Social welfare issues

Choice under endowment conditions

Lesson objectives: Master several expansion models and analysis methods of consumer theory.

5.1 Gifted Choice

5.2 Price Effects of Endowment Choice

5.3 Exchange

5.4 Intertemporal Selection

5.5 Uncertainty Selection

Producer Choice

Lesson objectives: Understand how producer selection is formalized and master the tools and models for analyzing producer behavior.

6.1 Technology and Production Function

6.2 Technical Features

6.3 Profit Maximization

6.4 Cost Minimization

6.5 Returns to Scale and Cost Function

Perfectly competitive market

Lesson Objective: Understand the equilibrium and characteristics of a perfectly competitive market.

7.1 Manufacturer Supply

7.2 Industry Supply

7.3 General Equilibrium with Production

7.4 Comparative Advantage and Exchange



monopoly

Lesson objective: Master the classic model of monopolistic behavior.

8.1 Monopoly Enterprise Decision-Making

8.2 Monopoly and Natural Monopoly

8.3 Price Discrimination

8.4 Second-Degree Price Discrimination

8.5 Two-part charging



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Personal Center



Lesson objective: Master the classic oligopoly competition model.

9.1 Cournot Competition

9.2 Production Leadership

9.3 Price Competition

9.4 Market Collusion

Market failure

Lesson objectives: Understand the concepts of externalities and public goods, equilibrium and government governance.

10.1 Externalities

10.2 Public Goods

Close

Prerequisites

- Advanced Mathematics
- Elementary Microeconomics

References

- Hal-Varian: Microeconomics: A Modern Perspective (9th edition), Shanghai Sanlian Bookstore, 2015 Year.
- Hal R. Varian: Intermediate Microeconomics: A modern Approach, (9th Edition), WW Norton & Company, 2014, (English version, alternative 1).
- Jiang Zhongyi: Basic Methods of Mathematical Economics 2006 Year 11 Published by Commercial Press.

Frequently Asked Questions

Q: What are the textbooks for this course?

A: The main reference textbook for this course is Hal-Fan Liyi's Microeconomics: A Modern Perspective (9th Edition), Shanghai Sanlian Bookstore, 2015. If you don't have the textbook, listen carefully to this lesson.

It is also okay to take relevant notes.

Q: What prerequisites are required for this course?

A: A foundation in advanced mathematics is recommended, and even better still, a foundation in elementary microeconomics. However, even without this foundation, this course should be accessible.

Q: Can I take this course directly without studying economics?

A: Absolutely.

Q: Can I take this course if I don't have a good foundation in mathematics?

A: It is recommended to have a certain mathematical foundation, especially the foundation of calculus.

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2.2Assumptions about the production function and its inputs in the Solow model

2.3Steady-state analysis of the Solow model

2.4Comparative static analysis of the Solow model steady state and its interpretation of reality

3

Chapter 3 Frontiers of Economic Growth Theory II: Infinite Horizon and Overlapping Generations Model

-

3.1Ramsey Model1

3.2Ramsey Model2

3.3Ramsey Model3

3.4Ramsey Model4

3.5Ramsey Model5

3.6Diamond Model1

3.7Diamond Model2

4

Chapter 4: Frontiers of Economic Growth Theory III: New Growth Theory

-

4.1Overview of the Theory of Endogenous Economic Growth

4.2Overview of the Theory of Endogenous Economic Growth

4.3Research and development model

4.4Research and development model

4.5Research and development model

4.6Learning by Doing Model

4.7AKModel

5

Chapter 5 Real Business Cycle Theory

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5.1The meaning and characteristics of the economic cycle

5.2Classification of Business Cycles and Exogenous Business Cycle Theory

5.3The beginnings of business cycle theory

5.4Typical endogenous business cycle theory

5.5Traditional Keynesian business cycle theory

5.6Business cycle theory in monetarism and neoclassical macroeconomics

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